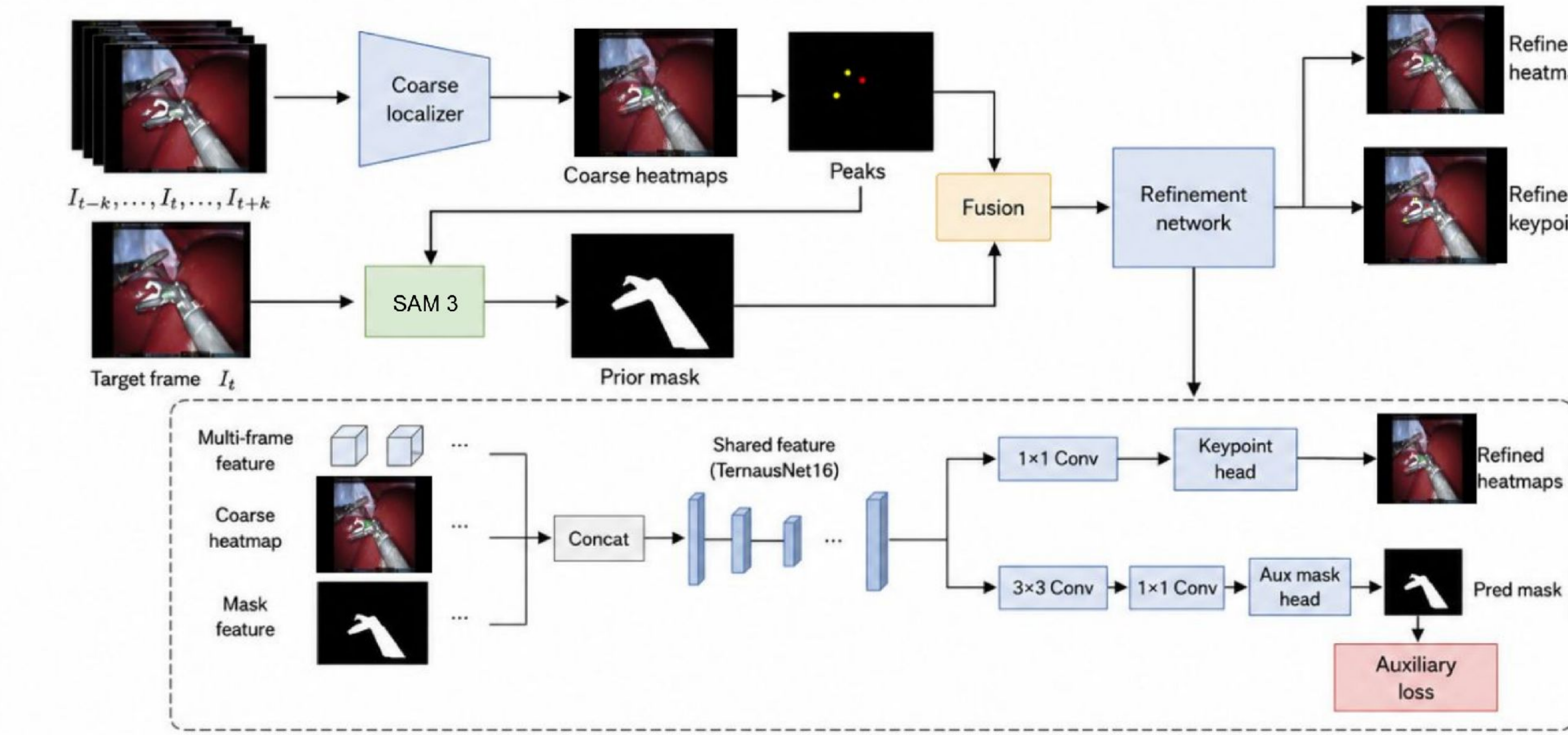


INTRODUCTION

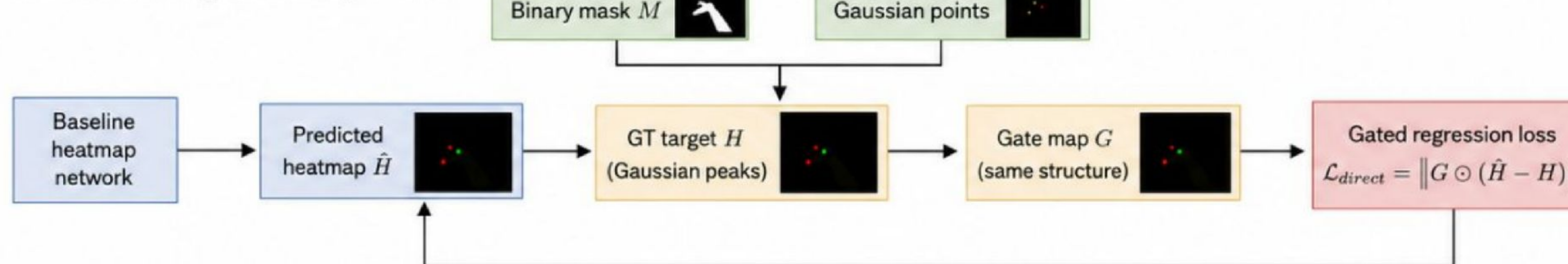
Tip and anchor landmarks drive pose estimation, motion analysis and skill assessment. They encode where the action happens, not just that a tool is present. Surgical video is messy. Smoke, bleeding, blur, several tools in frame. Foundation models now cope with all of that and hand you a clean instrument mask at no annotation cost. The mask is also where it stops. It covers the whole tool, while a functional landmark is one point on it, and which point depends on whether the surgeon is clipping, grasping or dissecting.

METHOD

(a) Prior-guided refinement with auxiliary mask supervision



(b) Direct mask-augmented supervision



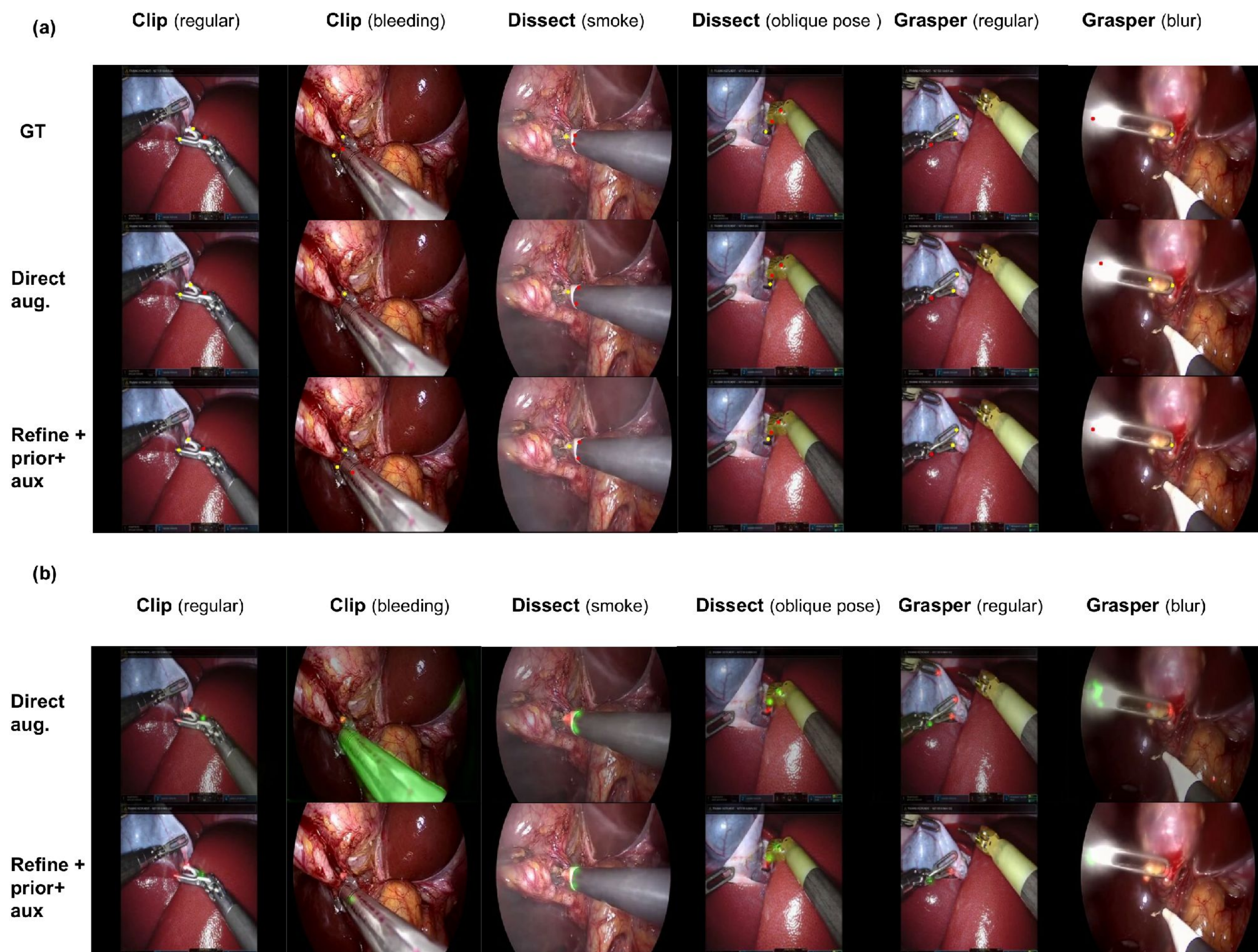
Direct mask augmented supervision (ablation)

SAM 3 masks reweight the heatmap regression in space. Structure goes straight into the target, and that turns out to be the wrong place for it.

Auxiliary supervision at the refinement stage

Same mask, different entry point. A parallel branch injects it earlier, so the shared feature carries the tool shape instead of the keypoint head.

QUALITATIVE RESULTS



Row (a) is keypoints against ground truth, row (b) is the heatmap behind them. Columns span bleeding, smoke, oblique tool pose and motion blur.

Coverage versus precision

Direct augmentation finds the instrument but spreads response across it. Prior guided refinement keeps one peak where the action is, and still uses the mask to get there.

QUANTITATIVE RESULTS

External comparison at a 15 px matching tolerance.

Model	Method	Tip localization				Anchor localization			
		P↑	R↑	F1↑	L2↓	P↑	R↑	F1↑	L2↓
<i>Clip</i>									
SimpleBaseline	Heatmap	93.3	25.5	40.1	8.88	70.1	17.8	28.8	11.61
YOLOv8-pose	Top-down	47.1	26.2	33.6	8.02	65.6	41.9	51.2	6.54
RTMPose	Top-down	50.2	50.2	50.2	12.73	49.3	49.4	49.4	9.07
Ours	Prior + aux refinement	77.9	48.4	59.7	9.05	61.5	51.1	55.8	10.19
<i>Grasp</i>									
SimpleBaseline	Heatmap	88.4	39.1	54.2	8.96	63.2	25.9	36.7	12.31
YOLOv8-pose	Top-down	52.5	31.5	39.4	10.52	60.4	43.4	50.5	11.50
RTMPose	Top-down	58.7	58.7	58.7	7.29	35.8	35.8	35.8	7.39
Ours	Prior + aux refinement	80.9	49.7	61.6	7.23	56.2	41.6	47.8	10.36
<i>Dissect</i>									
SimpleBaseline	Heatmap	46.4	26.4	33.6	14.97	78.2	39.9	52.8	10.49
YOLOv8-pose	Top-down	84.7	72.2	78.0	10.04	52.6	45.2	48.6	12.02
RTMPose	Top-down	69.3	68.8	69.1	8.90	59.1	59.4	59.2	8.28
Ours	Prior + aux refinement	81.1	72.6	76.6	10.47	67.6	52.2	58.9	8.49
<i>Overall</i>									
SimpleBaseline	Heatmap	53.1	28.5	37.1	13.38	77.3	39.4	52.2	10.58
YOLOv8-pose	Top-down	77.4	60.1	67.7	10.05	53.3	45.0	48.8	11.89
RTMPose	Top-down	65.5	65.2	65.4	8.70	57.0	57.2	57.1	8.25
Ours	Prior + aux refinement	81.0	65.5	72.4	9.80	66.6	51.3	58.0	8.64

DATASET

Action	YouTube	Cholec80	HeiChole	SurgVU	CRCD	Total
Clip	36	157	57	35	0	285
Dissect	611	3605	939	245	628	6028
Grasp	124	752	220	322	136	1554
Total	771	4514	1216	602	764	7867

7,867 clips · 60 videos · 5 sources. Split at the **video level**.

WHERE SHOULD STRUCTURE ENTER?

Method	LM	Clip		Grasp		Dissect		Overall			
		F1↑	L2↓	F1↑	L2↓	F1↑	L2↓	P↑	R↑	F1↑	L2↓
Coarse	Tip	53.4	9.02	60.5	7.40	51.7	10.44	61.1	46.7	52.9	9.92
	Anchor	45.9	9.30	46.5	10.49	51.8	8.26	67.3	41.7	51.5	8.36
Refine w/o prior	Tip	57.9	8.59	61.6	7.23	71.4	10.76	76.0	62.4	68.5	9.96
	Anchor	54.0	11.17	47.9	10.39	57.2	8.29	63.1	50.9	56.4	8.49
Direct mask-aug.	Tip	57.4	9.16	64.1	7.32	74.7	10.12	72.9	65.0	68.7	8.95
	Anchor	43.4	12.82	44.9	9.88	59.6	8.45	58.4	54.6	56.4	8.78
Refine + prior	Tip	58.5	7.98	61.6	7.21	75.7	10.65	80.3	64.7	71.6	9.87
	Anchor	63.8	10.69	47.5	10.16	58.7	8.71	66.3	51.4	57.9	8.85
Refine + prior + aux	Tip	59.7	9.05	61.6	7.23	76.6	10.47	81.0	65.5	72.4	9.80
	Anchor	55.8	10.20	47.8	10.36	58.9	8.49	66.6	51.3	58.0	8.64

- Refinement is the largest single gain; the prior adds **+3.1** on tips.

CONCLUSION

- This combination achieves the best performance: 72.4% Tip F1 and 58.0% Anchor F1, without manual pixel-level instrument-mask annotations.
- We introduce a prior-guided refinement framework that uses a prediction-derived SAM 3 mask as structural input and auxiliary mask supervision to guide landmark localization.
- In contrast, directly augmenting landmark heatmap targets with masks produces broad activations, revealing that dense structure should be used as guidance, not as the localization target.
- The mask prior provides instrument-level geometric context, while the auxiliary mask-prediction objective further regularizes the refinement features and yields the best overall localization performance.